

K -theory rules out an isomorphism between the Calkin algebra and its stable corona

Literature-implied answer to Question 32 in arXiv:2412.11191

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Status: literature-implied full answer. The answer is negative in ZFC, hence also under CH. The Calkin algebra has graded K -theory $(0, \mathbb{Z})$, whereas the corona of its stabilization has graded K -theory $(\mathbb{Z}, 0)$.

Source question

Głódkowski and Koszmider ask in Question 32, on source PDF page 15 [1]:

“Assume the continuum hypothesis CH. Are the algebras $\mathcal{Q}(\ell_2)$ and $\mathcal{Q}(\mathcal{Q}(\ell_2) \otimes \mathcal{K}(\ell_2))$ isomorphic?”

Here $\mathcal{Q}(A) = \mathcal{M}(A)/A$ denotes the corona algebra. Put

$$Q = \mathcal{B}(\ell_2)/\mathcal{K}(\ell_2), \quad C = \mathcal{M}(Q \otimes \mathcal{K})/(Q \otimes \mathcal{K}).$$

Supporting theorem and identification

For every C^* -algebra A , the stable multiplier algebra $\mathcal{M}(A \otimes \mathcal{K})$ has vanishing K -theory:

$$K_0(\mathcal{M}(A \otimes \mathcal{K})) = K_1(\mathcal{M}(A \otimes \mathcal{K})) = 0. \quad (1)$$

This is the standard stable-multiplier theorem; Thiel invokes it on supporting PDF page 5 and, in Corollary 3.3 and Example 3.4 on pages 5–6, applies the resulting six-term exact sequence specifically to $Q \otimes \mathcal{K}$ [2].

First apply the six-term sequence to the Calkin extension

$$0 \longrightarrow \mathcal{K} \longrightarrow \mathcal{B}(\ell_2) \longrightarrow Q \longrightarrow 0.$$

Since $\mathcal{B}(\ell_2) = \mathcal{M}(\mathcal{K})$ has zero K -theory by (1), while $K_0(\mathcal{K}) \cong \mathbb{Z}$ and $K_1(\mathcal{K}) = 0$, its boundary maps give

$$K_0(Q) = 0, \quad K_1(Q) \cong \mathbb{Z}. \quad (2)$$

This is also the familiar Fredholm-index computation of the K -theory of the Calkin algebra.

Now apply the six-term sequence to

$$0 \longrightarrow Q \otimes \mathcal{K} \longrightarrow \mathcal{M}(Q \otimes \mathcal{K}) \longrightarrow C \longrightarrow 0.$$

Stability of K -theory and (1) make both boundary maps isomorphisms of degree one:

$$K_0(C) \cong K_1(Q \otimes \mathcal{K}) \cong K_1(Q) \cong \mathbb{Z}, \quad K_1(C) \cong K_0(Q \otimes \mathcal{K}) \cong K_0(Q) = 0.$$

Consequently

$$(K_0(Q), K_1(Q)) = (0, \mathbb{Z}) \quad \text{but} \quad (K_0(C), K_1(C)) = (\mathbb{Z}, 0).$$

An isomorphism of C^* -algebras induces degree-preserving isomorphisms on K_0 and K_1 . Therefore Q and C are not isomorphic. The computation is a theorem of ZFC and is valid in every model, so CH is irrelevant.

Provenance and scope

The supporting paper predates the source question and does not state that it is answering Question 32. The answer is an agent-identified direct consequence of its cited stable-multiplier theorem and six-term computation, so the proper provenance is *literature-implied answer*. This fully resolves Question 32 but says nothing about the masa embedding questions in Question 33.

References

- [1] D. Głódkowski and P. Koszmider, *Products of C^* -algebras that do not embed into the Calkin algebra*, arXiv:2412.11191 (2024), Question 32.
- [2] H. Thiel, *Real rank of some multiplier algebras*, arXiv:2402.01022, Corollary 3.3 and Example 3.4.